Chapter 5
Matrix Factorization

We will investigate the matrix factorization problem: for a given square matrix $\mathbf{A}$ we are looking for special matrices $\mathbf{B}$ and $\mathbf{C}$ such that $\mathbf{A} = \mathbf{BC}$. First we study the LU factorization, and then the Cholesky factorization.

5.1. LU Factorization

Let $\mathbf{A}$ be an $n \times n$ matrix. The product $\mathbf{A} = \mathbf{LU}$ is called *LU factorization* of $\mathbf{A}$ or *Doolittle's method* if $\mathbf{L}$ is lower triangular with all entries 1 in the main diagonal, and $\mathbf{U}$ is upper triangular.

Theorem 5.1. *Let $\mathbf{A}$ be a nonsingular square matrix. If the LU factorization of $\mathbf{A}$ exists, then its is unique.*

Proof. Suppose $\mathbf{A} = \mathbf{L}_1\mathbf{U}_1 = \mathbf{L}_2\mathbf{U}_2$ are two LU factorizations of the matrix $\mathbf{A}$. Since $\det(\mathbf{A}) = \det(\mathbf{L}_1)\det(\mathbf{U}_1) = \det(\mathbf{L}_2)\det(\mathbf{U}_2) \neq 0$, therefore, $\mathbf{L}_1$, $\mathbf{L}_2$, $\mathbf{U}_1$ and $\mathbf{U}_2$ are nonsingular matrices. Hence $\mathbf{L}_2^{-1}\mathbf{L}_1 = \mathbf{U}_2\mathbf{U}_1^{-1}$. Using Theorem 3.6, the matrix $\mathbf{L}_2^{-1}\mathbf{L}_1$ is lower triangular, and the matrix $\mathbf{U}_2\mathbf{U}_1^{-1}$ is upper triangular. Therefore, both matrices are diagonal. It is easy to see that the main diagonal of $\mathbf{L}_2^{-1}\mathbf{L}_1$ consists of only 1 entry, hence $\mathbf{L}_2^{-1}\mathbf{L}_1 = \mathbf{U}_2\mathbf{U}_1^{-1} = \mathbf{I}$, which implies that $\mathbf{L}_1 = \mathbf{L}_2$ and $\mathbf{U}_1 = \mathbf{U}_2$. $\square$

Consider the definition of the Gaussian elimination introduced in Section 3.3. Let $l_{i1} = a_{i1}/a_{11}$, $i = 2, 3, \ldots, n$, as in Section 3.3, and define the lower triangular matrix

$$\mathbf{L}_1 := \begin{pmatrix} 1 & & & & \\ -l_{21} & 1 & & & \\ -l_{31} & & 1 & & \\ \vdots & & & \ddots & \\ -l_{n1} & & & & 1 \end{pmatrix},$$

where the missing elements are all equal to 0. It is easy to check whether the product $\mathbf{L}_1\mathbf{A}$ gives the matrix $\mathbf{A}^{(1)}$, the matrix obtained performing the first elimination step of the Gaussian elimination on the coefficient matrix: $\mathbf{A}^{(1)} = \mathbf{L}_1\mathbf{A}$. Similarly, let $l_{i2} = a_{i2}^{(1)}/a_{22}^{(1)}$, $i = 3, 4, \ldots, n$, and define the matrix

$$\mathbf{L}_2 := \begin{pmatrix} 1 & & & & \\ & 1 & & & \\ & -l_{32} & 1 & & \\ & \vdots & & \ddots & \\ & -l_{n2} & & & 1 \end{pmatrix},$$

where all elements in the main diagonal are 1, in the second column the elements under the diagonal are $-l_{32}$, $-l_{42}$, ..., $-l_{n2}$, and all the other elements are 0. Then $\mathbf{A}^{(2)} = \mathbf{L}_2\mathbf{A}^{(1)}$ holds. We define the lower triangular matrices $\mathbf{L}_3, \dots, \mathbf{L}_{n-1}$ in a similar manner. Simple computation shows

$$\mathbf{L}_{n-1}\mathbf{L}_{n-2}\cdots\mathbf{L}_1 = \begin{pmatrix} 1 & & & & \\ -l_{21} & 1 & & & \\ -l_{31} & -l_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ -l_{n1} & -l_{n2} & \cdots & -l_{n,n-1} & 1 \end{pmatrix}, \tag{5.1}$$

and

$$\begin{aligned} \mathbf{L} &:= (\mathbf{L}_{n-1}\mathbf{L}_{n-2}\cdots\mathbf{L}_1)^{-1} \\ &= \mathbf{L}_1^{-1}\cdots\mathbf{L}_{n-2}^{-1}\mathbf{L}_{n-1}^{-1} \\ &= \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & 0 & 1 & & \\ \vdots & 0 & \ddots & \ddots & \\ l_{n1} & 0 & \cdots & 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & & & & \\ 0 & 1 & & & \\ 0 & 0 & 1 & & \\ 0 & \vdots & \ddots & \ddots & \\ 0 & 0 & \cdots & l_{n,n-1} & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & & & & \\ l_{21} & 1 & & & \\ l_{31} & l_{32} & 1 & & \\ \vdots & \vdots & \ddots & \ddots & \\ l_{n1} & l_{n2} & \cdots & l_{n,n-1} & 1 \end{pmatrix}. \end{aligned} \tag{5.2}$$

Let $\mathbf{U} := \mathbf{A}^{(n-1)}$, i.e., the upper triangular matrix which is the result of the Gaussian elimination. Then $\mathbf{U} = \mathbf{L}_{n-1}\cdots\mathbf{L}_1\mathbf{A}$, which gives $\mathbf{A} = \mathbf{LU}$. We have proved the following result.

Theorem 5.2. *If the Gaussian elimination can be performed on a square matrix* $\mathbf{A}$*, then the LU factorization* $\mathbf{A} = \mathbf{LU}$ *exists. Then* $\mathbf{U}$ *is the upper triangular matrix obtained by the Gaussian elimination, and* $\mathbf{L}$ *is defined by (5.2), where* l_{ij} *denote the factors used in the Gaussian elimination.*

Example 5.3. Consider the coefficient matrix of Example 3.22:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix}.$$

As we saw in Example 3.22, the Gaussian elimination can be performed on $\mathbf{A}$, and $l_{21} = 2$, $l_{31} = -1$, $l_{41} = -2$, $l_{32} = 0$, $l_{42} = -1$ and $l_{43} = 6$. If we compute the LU factorization, then we write down the Gaussian elimination so that the factors l_{ij} can be written in place of the

elements which are eliminated (changed to 0):

$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -3 & 0 & -6 \end{pmatrix} \sim$$
$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -1 & 6 & 2 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & 3 & 6 & 8 \\ -1 & 0 & 1 & -6 \\ -2 & -1 & 6 & 38 \end{pmatrix}.$$

Now in the last matrix the elements in the main diagonal and above are the elements of the matrix $\mathbf{U}$, and the elements below the main diagonal are the entries of $\mathbf{L}$. Therefore,

$$\begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -2 & -1 & 6 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & -2 & -2 \\ 0 & 3 & 6 & 8 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 38 \end{pmatrix},$$

which can be checked by performing the product. □

The following results can be proved easily.

Theorem 5.4. *If all the principal minors of* $\mathbf{A}$ *are nonzero, then the Gaussian elimination can be performed without row changes, and so the LU factorization* $\mathbf{A} = \mathbf{LU}$ *exists.*

Theorem 5.5. *For any invertible square matrix* $\mathbf{A}$ *there exists a permutation matrix* $\mathbf{P}$ *such that the LU factorization* $\mathbf{PA} = \mathbf{LU}$ *exists.*

If an LU factorization $\mathbf{A} = \mathbf{LU}$ is known, then we can solve linear systems with the coefficient matrix $\mathbf{A}$ efficiently. Consider the system $\mathbf{Ax} = \mathbf{b}$. We introduce the new variable $\mathbf{y} = \mathbf{Ux}$. Then the original system is equivalent to

$$\begin{aligned} \mathbf{Ly} &= \mathbf{b} \\ \mathbf{Ux} &= \mathbf{y}, \end{aligned}$$

where both systems are triangular. We solve the first equation using a forward substitution method for $\mathbf{y}$, and then the second equation using the backward substitution method for $\mathbf{x}$. It is easy to check that $n^2 + \mathcal{O}(n)$ number of multiplications/divisions are needed to solve the two triangular systems, and to compute the LU factorization, $n^3/3 + \mathcal{O}(n^2)$ number of multiplications/divisions are needed. It is especially efficient if we solve several linear system with the same coefficient matrix.

Exercises

1. Compute the LU factorization of the following matrices:

(a) $\begin{pmatrix} 2 & 3 & -1 \\ -1 & -2 & -1 \\ 0 & 2 & 4 \end{pmatrix}$ (b) $\begin{pmatrix} 4 & -1 & 2 \\ -12 & 0 & -1 \\ 8 & -17 & 26 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 3 & -1 & 2 \\ -2 & -4 & 5 & -5 \\ 0 & 6 & 6 & -2 \\ 2 & 4 & -14 & 16 \end{pmatrix}$ (d) $\begin{pmatrix} 2 & -1 & 3 & -2 \\ -8 & 5 & -7 & 7 \\ 2 & -4 & -14 & 0 \\ -4 & 7 & 23 & 4 \end{pmatrix}$

2. Show that the matrix
$$\begin{pmatrix} 2 & 2 & 3 \\ 1 & 1 & 4 \\ 1 & 0 & 1 \end{pmatrix}$$
has no LU factorization.
3. Show that the matrix
$$\begin{pmatrix} 1 & 1 & -1 \\ 2 & 2 & 2 \\ 3 & 3 & -4 \end{pmatrix}$$
has infinitely many LU factorization. Do not we get a contradiction to Theorem 5.1?
4. Prove Theorem 5.4. (Hint: Use that during the elimination steps the principal minors of $\mathbf{A}^{(k-1)}$ and $\mathbf{A}^{(k)}$ are equal. Why?)
5. Prove Theorem 5.5.
6. Solve the linear systems given in Exercise 1 of Section 3.3 using LU factorization.

5.2. Cholesky Factorization

Let $\mathbf{A}$ be a symmetric matrix. The factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ of the matrix $\mathbf{A}$, where $\mathbf{L}$ is a lower triangular matrix, is called the *Cholesky factorization.*

We note that if the Cholesky factorization exists, it is not unique. The next theorem formulates a sufficient condition for the existence of the Cholesky factorization.

Theorem 5.6. *If $\mathbf{A}$ is positive definite, then the Cholesky factorization $\mathbf{A} = \mathbf{L}\mathbf{L}^T$ exists, the matrix $\mathbf{L}$ is real, and we can select positive elements in the main diagonal of $\mathbf{L}$.*

Proof. We prove the statement using mathematical induction with respect to the dimension of the matrix $\mathbf{A}$. The statement is obvious for 1×1 matrices. Suppose the statement of the theorem holds for $(n-1) \times (n-1)$ matrices, and let $\mathbf{A}$ be an $n \times n$ matrix. We partition the matrix $\mathbf{A}$ in the following form:
$$\mathbf{A} = \begin{pmatrix} \mathbf{X} & \mathbf{y} \\ \mathbf{y}^T & a_{nn} \end{pmatrix},$$
where $\mathbf{X}$ is an $(n-1) \times (n-1)$ matrix, $\mathbf{y}$ is an $n-1$-dimensional column vector. Theorem 3.10 yields that $\mathbf{X}$ is positive definite. We are looking for the Cholesky factorization of $\mathbf{A}$ in the form
$$\mathbf{A} = \begin{pmatrix} \mathbf{X} & \mathbf{y} \\ \mathbf{y}^T & a_{nn} \end{pmatrix} = \begin{pmatrix} \tilde{\mathbf{L}} & \mathbf{0} \\ \mathbf{c}^T & d \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{L}}^T & \mathbf{c} \\ \mathbf{0}^T & d \end{pmatrix}. \tag{5.3}$$
Here $\tilde{\mathbf{L}}$ is an $(n-1)\times(n-1)$ dimensional lower triangular matrix, $\mathbf{c}$ is an $n-1$-dimensional column vector, $d \in \mathbb{R}$. If we perform the matrix multiplication on the partitioned matrices, we get the relations
$$\mathbf{X} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^T, \qquad \tilde{\mathbf{L}}\mathbf{c} = \mathbf{y} \quad \text{and} \quad \mathbf{c}^T\mathbf{c} + d^2 = a_{nn}.$$
By the induction hypothesis the equation $\mathbf{X} = \tilde{\mathbf{L}}\tilde{\mathbf{L}}^T$ has a lower triangular solution $\tilde{\mathbf{L}} \in \mathbb{R}^{(n-1)\times(n-1)}$, where in the main diagonal we can select positive elements. This yields

that $\tilde{\mathbf{L}}$ is nonsingular, so the equation $\tilde{\mathbf{L}}\mathbf{c} = \mathbf{y}$ has a unique solution $\mathbf{c}$. Let d be a (possibly complex) root of the equation $\mathbf{c}^T\mathbf{c} + d^2 = a_{nn}$. Then relation (5.3) holds. d can be selected to be a positive real if and only if $d^2 = a_{nn} - \mathbf{c}^T\mathbf{c} > 0$. Relation (5.3) implies $\det(\mathbf{A}) = \det(\tilde{\mathbf{L}})^2 d^2$. Since $\mathbf{A}$ is positive definite, it follows $\det(\mathbf{A}) > 0$ (see Theorem 3.10). This yields that d^2 is positive, hence d can be selected to be a positive real. □

Example 5.7. Find the Cholesky factorization of the matrix

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix}.$$

We write

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{pmatrix} \begin{pmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{pmatrix}$$

We consider first the equation for the first row first element: $4 = l_{11}^2$. This can be solved for l_{11}: the positive solution is $l_{11} = 2$. Then we consider the elements under the main diagonal of the first column: $-8 = l_{21}l_{11}$, $4 = l_{31}l_{11}$. These can be solved uniquely for l_{21} and l_{31}: $l_{21} = -4$, $l_{31} = 2$. Now we consider the element of the main diagonal of the second column: $17 = l_{21}^2 + l_{22}^2$. Its positive solution is $l_{22} = 1$. Then look at the element in the second column under the main diagonal: $-11 = l_{31}l_{21} + l_{32}l_{22}$. This can be solved as $l_{32} = -3$. Finally, the element in the third row and third column is $22 = l_{31}^2 + l_{32}^2 + l_{33}^2$. This gives $l_{33} = 3$. We have then

$$\begin{pmatrix} 4 & -8 & 4 \\ -8 & 17 & -11 \\ 4 & -11 & 22 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ -4 & 1 & 0 \\ 2 & -3 & 3 \end{pmatrix} \begin{pmatrix} 2 & -4 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 3 \end{pmatrix}.$$

□

We can generalize the method of the previous example:

Algorithm 5.8. Cholesky factorization

INPUT: $\mathbf{A}$
OUTPUT: $\mathbf{L}$

$l_{11} \leftarrow \sqrt{a_{11}}$
for $i = 2, \ldots, n$ **do**
  $l_{i1} \leftarrow a_{i1}/l_{11}$
end do
for $j = 2, \ldots, n-1$ **do**
  $l_{jj} \leftarrow \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}$
  for $i = j+1, \ldots, n$ **do**
    $l_{ij} \leftarrow \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik}l_{jk}\right)/l_{jj}$
  end do
end do
$l_{nn} \leftarrow \sqrt{a_{nn} - \sum_{k=1}^{n-1} l_{nk}^2}$
output$(l_{ij}, \quad i = 1, \ldots, n,\ j = 1, \ldots, i)$

The operation count of Algorithm 5.8 is $n^3/6+n^2/2-2n/3$ number of multiplications and divisions, and $n^3/6-n/6$ number of additions and subtractions, and n number of square roots.

Exercises

1. Compute the Cholesky factorization of the following matrices:

$$\text{(a)} \begin{pmatrix} 16 & -8 & -12 \\ -8 & 8 & 4 \\ -12 & 4 & 35 \end{pmatrix}, \qquad \text{(b)} \begin{pmatrix} 4 & -2 & -4 \\ -2 & 26 & 7 \\ -4 & 7 & 6 \end{pmatrix},$$

$$\text{(c)} \begin{pmatrix} 1 & -1 & -2 & 1 \\ -1 & 10 & 2 & 2 \\ -2 & 2 & 29 & 8 \\ 1 & 2 & 8 & 7 \end{pmatrix}, \qquad \text{(d)} \begin{pmatrix} 16 & -8 & 0 & -4 \\ -8 & 5 & 1 & 3 \\ 0 & 1 & 10 & -5 \\ -4 & 3 & -5 & 7 \end{pmatrix}.$$

2. Give an example for a matrix for which the Cholesky factorization is not unique.
3. Show that the matrix
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
has no Cholesky factorization.
4. Prove that the operation count of Algorithm 5.8 is $n^3/6+n^2/2-2n/3$ number of multiplications and divisions, and $n^3/6-n/6$ number of additions and subtractions, and n number of square roots.
5. Show without using Theorem 3.10 that the matrix $\mathbf{X}$ in the proof of Theorem 5.6 is positive definite.